AI-Powered Real Estate Property Valuation Assistant

End-to-End System Architecture • Author / Presenter: Kunjal Ahuja • Prodapt Capstone Project

1. DATASET

Ames Housing Dataset

Total: 10,000 Records
• 2,930 Real Sales (OpenML 43926)
• 7,070 Synthetic Records (Copula)
28 Neighborhoods
Year Built: 1872 – 2010
15 Key Physical Fields

2. PREPROCESSING

Input Guardrails & ETL

15 Key Fields Standardized
1-10 Ordinal Quality Scale
Derived: Age, EffAge, PPSF
Guardrails: 14 Fields Validated
100 <= SqFt <= 15,000
Prompt Injection Defense

3. CHUNKING

Document Serialization

1 Property = 1 Document
No Splitting / No Overlap
Searchable Text Payload:
Location, SqFt, Beds, Baths
Year Built & Remodeled
Price & Historical PPSF

4. EMBEDDINGS

Dense Representation

SentenceTransformers
Model: all-MiniLM-L6-v2
Dimension: 384-d Dense
Local Offline Inference
Cached Unit Vectors (.npy)
Sub-millisecond Query

5. VECTOR DATABASE

Vector Storage Layer

In-Memory NumPy Index
embeddings.npy Matrix
Shape: (10,000 x 384)
Size: ~15.3 MB in Memory
Sub-5ms Cosine Similarity
Zero Database Overhead

6. RETRIEVER

True Hybrid Search Engine

Candidate TF-IDF Matrix
Dense Cosine Similarity
Fusion: 0.55*Dense + 0.45*TFIDF
Comparability Index:
Location 30%, SqFt 25%, Qual 20%
Age 15%, Room Count 10%
Comps Filter: Exclude Synthetic
Dynamic Neighborhood Widening

7. LLM / RAG

Grounded Retrieval Synthesis

Dual-Mode Operation:
• Live: OpenAI gpt-4o-mini
• Offline: Heuristic Engine
Grounded Valuation Narrative
Cites Verified Comps & PPSF
Zero Financial Hallucination
Pydantic Schema Validation
DeepEval Benchmark Integration

8. AGENTS

Multi-Agent Collaborative System Core

Collaborative Multi-Agent Architecture:
• Classification Agent: Market Tier, Typology, Condition Grade, Demographics
• Supervised Regressor: GradientBoostingRegressor (Test R^2 = 0.9198, RMSE = $25,361.84, MAPE = 8.87%)
Honest Real-Only Test Set (n=586, random_state=42) + 90% Empirical Prediction Intervals
• Basic Valuation Agent: Median Comparable Price ($) + Size-Weighted PPSF Valuation
• Dynamic Reconciliation: comp_fit weighting (comp_weight: 0.1–0.5, ml_weight: 0.5–0.9)
• Property Risk Agent: 0-100 Risk Score (Physical Age, Condition Gap, Volatility, Liquidity)
• Feature Impact Agent: Neighborhood Location Premiums & What-If Renovation Simulator
• LLM-as-Judge Auditor: Multi-criterion valuation audit issuing formal verdict (APPROVED)

9. API LAYER

FastAPI REST Microservice Layer

16 REST Endpoints (Base /api/v1):
GET /, GET /health, POST /guardrails/validate
POST /search, POST /classify, POST /valuation/basic
POST /valuation/predict-price, POST /valuation/predict-risk
POST /valuation/impact-analysis, POST /valuation/collaborate
POST /valuation/judge-validate, POST /feedback/submit
GET /feedback/summary, GET /evaluation/run, GET /evaluation/history
GET /dataset/info • OpenAPI Swagger UI at /docs

10. USER INTERFACE

React 18 + Vite Production Frontend

Modern SPA Architecture (frontend/):
• Left Sidebar: Shared 14-Field Property Input Form
• 7 Dedicated Accessible Workflow Tabs:
1. Valuation (RangeBar, 2-Sentence Plain-English Rationale)
2. Comparable Sales (Verified Historical Transactions)
3. ML Prediction (Gradient Boosting Point & 90% Interval)
4. What-If Simulator (Interactive Renovation Value Lift)
5. Risk & Audit (0-100 Multi-Factor Risk & Judge Verdict)
6. Feedback (Appraiser Rating & Bias Calibration)
7. Evaluation (DeepEval Quality Benchmark Runner)
Theme: Dark/Light Mode with System Default & LocalStorage

★. EVALUATION & FEEDBACK

Continuous Learning & DeepEval Benchmark

DeepEval v4.2.8 Quality Suite (5 Scenarios):
• 100% Pass Rate (5/5 Passed in 61.98s live run)
• Faithfulness: 0.900 | Answer Relevancy: 0.880
• Context Precision: 0.850 | Context Recall: 0.850
• Valuation Accuracy: 1.000
Continuous Improvement Feedback Loop:
• Logs Appraiser Ratings & Verified Closing Prices
• Uncalibrated Bias Tracking (Prevents Drift)
• Dynamic Calibration Factor Applied in Real-Time